

MCB

Inventory Control System

Full Workflow & Responsibility Specification

Purpose	PD review and workflow design freeze
Baseline	MCB Inventory System FINAL v94
Working candidate	v95 workflow-lock candidate
Review date	28 August 2026

Prepared as a management-ready summary plus implementation control specification. Unresolved business authority is explicitly marked NEEDS DECISION rather than guessed.

Document control

This document is the controlling business design for the final MCB Inventory System candidate. Coding should not silently change a workflow marked LOCKED. Anything unresolved remains NEEDS DECISION until approved.

LOCKED	NEEDS DECISION	BUILD GAP	FUTURE PHASE
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Design principles

- One central multi-site, multi-user system.
- Global Item Master with site-scoped stock positions.
- Current Stock is controlled/derived; no manual overwrite of posted quantity.
- On Hand = physical quantity; Reserved = approved transfer commitment; Available = On Hand - Reserved.
- Approved BOM is a hard operational limit for BOM-controlled Stock In/Out.
- Posted history is immutable; corrections/reversals preserve the original.
- Role + Site Assignment + Permission Overrides + Active/Inactive is the target access model.
- Normal flow is fast; exceptions receive human control.

Detailed workflows

1. Item Master

LOCKED

Control	Definition
Trigger	Need a new controlled item identity.
Creator / Data Entry	Authorized master-data user.
Checker	Master-data checker per configured permission.
Approver / Control	Policy if item approval is required.
Exception Owner	Developer for duplicates/data conflict.
CC / Informed	Relevant inventory/QS users.
System Control	Validate unique Item Code, type/category/UOM/status. Do not create zero stock position at every site.
Stock Effect	None.
BOM Effect	None until BOM/cost-code link.
Evidence	Item definition/photo/specification as applicable.
Audit Record	Create/edit/status user + timestamp + before/after.

2. Opening Balance

LOCKED DIRECTION

Control	Definition
Trigger	Site/item initialization before normal operation.

Creator / Data Entry	Authorized site user.
Checker	Stock Controller.
Approver / Control	Stock Controller.
Exception Owner	Stock Controller.
CC / Informed	Site team.
System Control	Only initialization; subsequent changes use Adjustment.
Stock Effect	Increase/decrease only after approval.
BOM Effect	Pre-existing stock BOM treatment NEEDS DECISION.
Evidence	Opening reference/count evidence.
Audit Record	Requester, approver, before/after, reason, timestamp.

3. Stock In

LOCKED CORE

Control	Definition
Trigger	Supplier delivery reaches site/warehouse.
Creator / Data Entry	Site Stockkeeper checks material + DN and submits.
Checker	Stock Controller verifies DN/evidence and accepted quantity.
Approver / Control	Stock Controller posts verified receipt.
Exception Owner	QS for BOM Hold; supplier/site for delivery exception.
CC / Informed	Site/management through status/reporting.
System Control	Validate site/item/cost code/supplier/evidence/BOM/duplicate. Stockkeeper submission stays Pending Verification.
Stock Effect	On Hand increases only on verification/post by accepted quantity.
BOM Effect	Purchase against Approved BOM; excess is blocked.
Evidence	DN, photo/evidence, reference, supplier, accepted/rejected qty.
Audit Record	Transaction Date, Posted At, creator, verifier, evidence, status history.

4. Daily Stock Out Batch

LOCKED CORE / BUILDING

Control	Definition
Trigger	Daily paper warehouse issue sheet.

Creator / Data Entry	Stock Controller in Phase 1.
Checker	System validates whole batch before post.
Approver / Control	No per-line approval in normal flow.
Exception Owner	QS for BOM Hold; Stock Controller for source-sheet correction.
CC / Informed	Site/management via records.
System Control	Batch/grid; aggregate duplicate lines; Qty <= Available and Qty <= BOM Remaining.
Stock Effect	All valid rows post together; no stock effect while drafting.
BOM Effect	Consumption recorded per site/item/cost code.
Evidence	Daily sheet ref, issued-to/responsible, work/remark.
Audit Record	Physical Transaction Date + Posted At/By + batch/line trace.

5. Site Transfer

LOCKED CORE / BUILD GAP

Control	Definition
Trigger	One site needs stock from another site.
Creator / Data Entry	Sender-side authorized user.
Checker	System availability; receiver checks actual delivered.
Approver / Control	Transfer approval owner NEEDS DECISION.
Exception Owner	Sender/receiver normal mismatch; Stock Controller serious/unresolved.
CC / Informed	Both involved sites only.
System Control	Draft -> Pending -> Approved/Reserved -> Dispatch -> In Transit -> Receive -> Complete/Disputed.
Stock Effect	Approval reserves; dispatch reduces sender On Hand; receipt adds receiver actual.
BOM Effect	Transfer allocation timing NEEDS DECISION.
Evidence	Transfer form, sender/receiver evidence, expected receipt.
Audit Record	One Transfer ID, actor/time/status history, sent/received/disputed qty.

6. Material Return

LOCKED DIRECTION / BUILD GAP

Control	Definition
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Trigger	Unused consumable returned to warehouse.
Creator / Data Entry	Authorized site user.
Checker	Warehouse checker.
Approver / Control	Per permission.
Exception Owner	Stock Controller if quantity/condition disputed.
CC / Informed	Site.
System Control	Must link to original Stock Out; cannot return more than eligible issued qty.
Stock Effect	Accepted return increases On Hand.
BOM Effect	Reduces net consumption while gross history remains.
Evidence	Original issue, condition, return reason.
Audit Record	Original transaction link, before/after, acceptor.

7. Equipment / Reusable Tools

LOCKED DIRECTION / BUILD GAP

Control	Definition
Trigger	Reusable item leaves warehouse custody.
Creator / Data Entry	Authorized site user/warehouse.
Checker	Borrower/warehouse checks condition.
Approver / Control	Per permission.
Exception Owner	Lost/Damaged/Repair/Scrap authority NEEDS DECISION.
CC / Informed	Assigned site/management exception view.
System Control	Checkout -> Outstanding -> Return; Good/Damaged/Lost condition.
Stock Effect	Custody/availability state rather than consumable issue.
BOM Effect	Generally not consumable usage; project control policy as configured.
Evidence	Borrower, checkout/return time, condition, evidence.
Audit Record	Full custody/condition event history.

8. Physical Count

LOCKED CORE / BUILD GAP

Control	Definition
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Trigger	Weekly/monthly control cycle.
Creator / Data Entry	Weekly: Stockkeeper + Stock Controller. Monthly: + Admin.
Checker	Joint count and variance recount.
Approver / Control	Final variance adjustment authority NEEDS DECISION.
Exception Owner	Stock Controller prepares/resolves variance.
CC / Informed	Admin/management per monthly process.
System Control	Blind count; 1st Count -> Recount variances -> Final Count; snapshot/cutoff.
Stock Effect	No direct overwrite; only controlled adjustment after decision.
BOM Effect	No direct BOM effect unless correction policy requires trace.
Evidence	Physical Count Form, count round, condition/remark.
Audit Record	Count session, snapshot, counters, rounds, variance and linked adjustment.

9. Correction / Reversal

LOCKED

Control	Definition
Trigger	A posted movement is discovered to be wrong.
Creator / Data Entry	Authorized controller.
Checker	System/audit checks original link and reason.
Approver / Control	Sensitive authority per permission.
Exception Owner	Stock Controller/Developer.
CC / Informed	Audit/report consumers.
System Control	Draft editable; Posted immutable; no delete. New linked reversal/correction only.
Stock Effect	New audited delta changes stock.
BOM Effect	Recompute valid usage/purchase history without erasing original.
Evidence	Original transaction, correction reference/reason.
Audit Record	Original ID, before/change/after, corrected by/time.

10. Month Close / Backdating

LOCKED CONCEPT / BUILD GAP

Control	Definition
Trigger	Period-end control.
Creator / Data Entry	Authorized posting user may backdate physical event only in open period.
Checker	System validates period status.
Approver / Control	Close/reopen authority NEEDS DECISION.
Exception Owner	Developer for technical exception.
CC / Informed	Management/finance as required.
System Control	Closed period blocks normal posting; later fix uses controlled correction/reversal.
Stock Effect	No direct close effect; it restricts posting.
BOM Effect	Preserves period reporting.
Evidence	Close checklist/report.
Audit Record	Closed/reopened by/time/reason.

11. Site Handover / Warranty / Closure

LOCKED / BUILD GAP

Control	Definition
Trigger	Construction progresses to handover and defect/warranty period.
Creator / Data Entry	Site/Admin team.
Checker	Stock Controller and equipment/transfer owners clear open records.
Approver / Control	Final closure authority NEEDS DECISION.
Exception Owner	Site/Admin/Stock Controller.
CC / Informed	Management.
System Control	Active Construction -> Handover -> Warranty -> Final Closure -> Archived. History never deleted.
Stock Effect	Final closure requires cleared stock/equipment/transfers/open transactions.
BOM Effect	BOM/report history retained.
Evidence	Closure checklist/handover evidence.
Audit Record	Lifecycle status history and closure decision.

12. User Access

LOCKED ARCHITECTURE / PARTIAL BUILD

Control	Definition
Trigger	User needs system access or access change.
Creator / Data Entry	Developer.
Checker	Developer.
Approver / Control	Developer.
Exception Owner	Developer.
CC / Informed	Audit/management as required.
System Control	Role -> Site Assignment -> Permission Overrides -> Active/Inactive. Explicit Deny wins. No duration/expiry.
Stock Effect	None.
BOM Effect	None.
Evidence	Access request/business authorization.
Audit Record	Every role/site/override/status change logged.

Responsibility matrix

Workflow	Do / Create	Check	Approve / Control	Exception Owner	Stock timing
Item Master	Master-data user	Configured checker	Configured authority	Developer	None
Opening Balance	Site user	Stock Controller	Stock Controller	Stock Controller	On approval
Stock In	Stockkeeper	Stock Controller	Stock Controller	QS / supplier-site exception	On verification
Stock Out	Stock Controller	System batch validation	Normal: none per line	QS on BOM Hold	On batch post
Transfer	Sender user	Receiver actual check	TBD	Stock Controller serious mismatch	Reserve/dispatch/receipt
Equipment	Site/warehouse	Borrower/warehouse	Per permission	TBD lost/damaged	Custody state
Weekly Count	Stockkeeper + Stock Controller	Joint	Variance TBD	Stock Controller	Adjustment only
Monthly Count	+ Admin	Joint	Variance TBD	Stock Controller	Adjustment only
Correction	Authorized controller	System/audit	Per permission	Stock Controller/Developer	New linked delta
BOM Revision	QS	Cost/control review	TBD	QS	No direct stock
Month Close	TBD	System checklist	TBD	Developer technical	Restricts posting
User Access	Developer	Developer	Developer	Developer	None

Role and permission baseline

Capability	Developer	Stock Controller	Stockkeeper	QS	Admin	Management
All-site access	Yes	Assigned	Assigned	Assigned	Configured	View/configured
Item Master	Full	View	View	View	Admin support	View
Current Stock	Full	Control/view	View	View	View	View
Stock In submit	Yes	Yes/post	Yes/pending	No	No default	No
Stock In verify	Yes	Yes	No	No	No default	No
Daily Stock Out	Yes	Yes	No default	No	No default	No
BOM edit/revision	Yes	No	No	Yes	No default	View
Transfer operations	Yes	Assigned-site	Assigned-site	View as needed	View/admin	View
Physical Count	Yes	Yes	Entry	View	Monthly joint	View
Corrections	Yes	Controlled	No	No	No default	No

User Access	Yes	No	No	No	No self-escalation	No
Reports/Audit	Yes	Operational	Operational	BOM/operational	Admin	Management

Permission Overrides are the target for exceptional users. Developer may Allow/Deny specific capabilities; explicit Deny wins. The server must enforce the same rule as the UI.

System controls and technical architecture

- Server-side authorization for every write; hiding a button is not security.
- Atomic posting: validate stock/BOM, write immutable transaction, update balance/reservation, workflow status and audit in one database transaction.
- Idempotency/duplicate protection for retries and double clicks.
- Relational ledger target for production multi-site operation. Whole-state revision storage is acceptable only as a transitional prototype mechanism.
- Site-scoped operational data; unrelated sites cannot act on a transfer.
- Backup is runtime-data recovery plus tested restore procedure, not only source-code ZIP.

PD decision register

ID	Decision	Recommended direction	Status
D1	BOM revision final approver	QS prepares; independent authorized approver confirms increase	NEEDS DECISION
D2	Transfer BOM timing	Receiver allocation at confirmed receipt; sender allocation release at controlled dispatch/receipt policy	NEEDS DECISION
D3	Count variance final authority	Independent control authority above/alongside preparer for material variance	NEEDS DECISION
D4	Month close/reopen authority	Limited Admin/Finance/Developer combination; reopen requires reason/audit	NEEDS DECISION
D5	Transfer approval owner	Choose role by site operating policy; avoid Stock Controller on every low-risk transfer if burnout risk	NEEDS DECISION
D6	PO/DN exceptions	Define no-PO, substitution, over-delivery, cancellation and rejected quantity	NEEDS DECISION
D7	Equipment exception authority	Define Lost/Damaged/Repair/Scrap/replacement conclusion	NEEDS DECISION

End-to-end acceptance scenario

The final workflow is considered locked only after the following chain is tested without bypass or hidden balance changes:

Item Master → BOM → Opening Balance → PO/DN → Stock In → Current Stock → Transfer → Stock Out → Material/Equipment Return → Physical Count → Adjustment → Month Close → Warranty → Final Closure

Production acceptance gate

- Two simultaneous users cannot consume the same last Available quantity.
- Stockkeeper Pending Verification Stock In cannot change Current Stock before verification.
- Repeated verification or request retry cannot double-post.
- Batch Stock Out aggregates duplicate item/cost-code rows before checking stock/BOM.
- Transfer partial receipt keeps unresolved quantity In Transit - Disputed.
- Count starts blind and recounts variance before adjustment.
- Posted records cannot be deleted; correction/reversal remains linked.
- Closed month rejects normal backdated entry.
- Archived/closed site history remains reportable.
- Role/site/permission isolation is tested by API, not only UI.